DATA 412/612 · Statistical Programming in R · Week 8

Lab 8: Relational Data

Join flights to airports, weather days, and planes.

Work in a **Quarto** document. Do **8.1** and **8.2** first.

```
library(tidyverse)
library(nycflights13)
```

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1. The five tables

Load `flights`, `planes`, `airlines`, `airports`, and `weather`. For each table, name a primary key (or say that there is none). A primary key is unique and never missing. It may be several columns. A foreign key points at another table's primary key and may repeat. Check uniqueness with `count()` and `filter(n > 1)`, and check for missing values. If rows come back, say whether those columns are not a key or are a key with an exception.

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2. Destinations and airports

1. How many distinct `dest` values are in `flights`?
2. Destinations that do not appear in `airports` (`anti_join` with `dest = faa`). How many flights is that?
3. Airports that never appear as a destination.
4. The ten most common destinations: FAA code, airport name, and flight count.

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3. Delay by timezone

Join `airports` onto `flights` by destination. Mean `arr_delay` by `tzone`. Plot it.

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4. Worst delay days

The ten days with the highest median `dep_delay`. Then `semi_join()` to keep every flight on those days.

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5. Plane age and delay

Join `planes`. Age is `2013 - year` of manufacture. Plot age against `dep_delay` (`geom_hex()` or `geom_smooth()`).

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6. Distance from NYC

1. Mean longitude and latitude of the three NYC airports (`EWR`, `JFK`, `LGA`). Treat that as the location of NYC.
2. Write a function `dist_nyc(lon, lat, nyclon, nyclat)` that returns great-circle distance from NYC using `geosphere::distm()`. What unit does `distm()` use?
3. Add that distance to every row of `airports`. Is mean arrival delay at a destination associated with distance from NYC?

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7. Fair-weather takeoffs

For each `tailnum`, the proportion of takeoffs in fair weather: `precip == 0`, `wind_speed < 20`, and `visib >= 10`. Join `weather` on `origin`, `year`, `month`, `day`, and `hour` (`hour` from `dep_time %/% 100`). Is that proportion associated with plane age?